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Descriptive Analytics Report

YouTube Content Performance at Scale

Note on data: this report is built on an uploaded video-performance dataset of 5,000 individual videos spanning 14 content categories, uploaded between January 2023 and August 2026, with recorded views, likes, comments, and channel subscriber counts at the video level. It is designed to show exactly what a descriptive analytics pipeline processing a platform's daily video-performance data would produce — the same structure applies once a full, continuously-refreshed feed is substituted in.

Table of Contents

1 Objective.....3

2 How this Would Work at Real Scale.....4

3 Upload and viewership trend.....5

4 Top performing videos.....6

5 Engagement Rate Distribution.....7

6 Category-Wise Demand.....8

7 Channel Reach Status.....9

8 Summary of Key Matrices.....10

9 Next Step 11

1. Objective

Build a descriptive analytics layer over video performance data to answer five recurring content and channel-strategy questions:

1. What is the top-performing videos, and which categories drive them?
2. How is viewership trending over time, upload by upload?
3. How is audience engagement distributed, and where is it slipping?
4. Where is viewer demand concentrated across content categories?
5. Which videos carry the highest breakout or engagement risk relative to their channel size?

2. How This Would Work at Real Scale

Processing video-performance data across an entire platform is an engineering problem before it's analytics one. A realistic architecture:

Layer	Purpose	Typical tools
Ingestion	Capture each view, like, comment, and subscribe event in real time	YouTube Data/Analytics
Stream processing	Roll up raw events into per-video, per-category, per-hour aggregates	-
Storage	Hold both raw events (for audits) and pre-aggregated tables (for speed)	-
Batch aggregation	Nightly rollups: daily views, engagement-rate averages, subscriber-tier reconciliation	-
Serving/BILAYER	Dashboards refreshed hourly or daily	Jupityer notebook

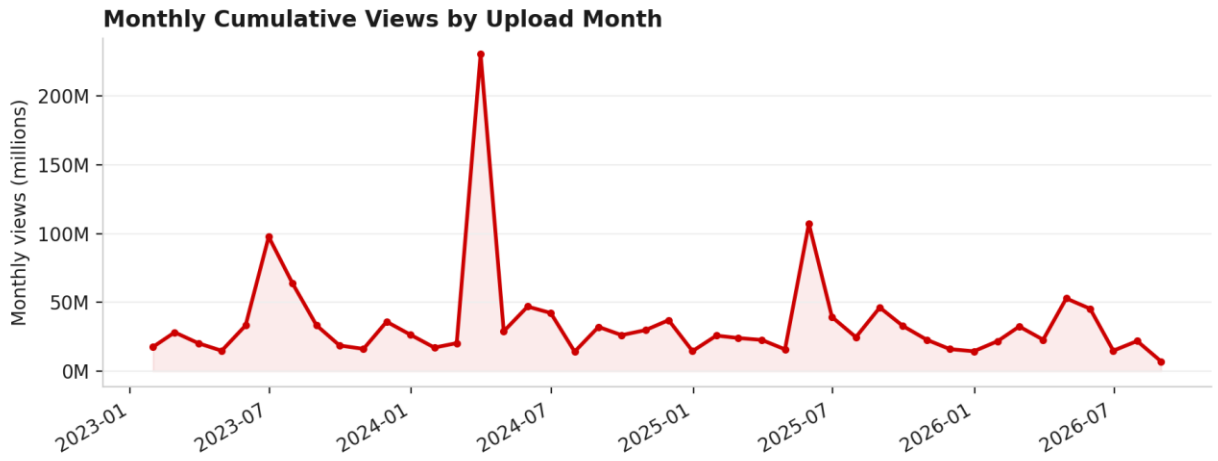
3. Upload & Viewership Trend

Across the dataset's 43-month window (January 2023 – August 2026): **5,000** videos were tracked, generating ~**1.56** billion cumulative views, at an average of ~**113** uploads and ~**36M** views per month, with sharp month-to-

month spikes driven by individual breakout videos rather than steady seasonal growth (e.g. March 2024 and May 2025 each cleared 100M monthly views on the strength of one or two outlier uploads).

KPI definitions used:

- Monthly views = sum of lifetime views across all videos uploaded that month
- Monthly upload volume = count of videos uploaded that month
- Because views accrue after upload rather than on a fixed daily cycle, month is used as the trend unit instead of day — the same daily grain from the reference report would apply directly to a live event-level feed



The graph shows that monthly views are highly fluctuating rather than steadily increasing. The biggest spike is approximately **230** million views, while most months remain considerably lower. This suggests that specific uploads or content releases can generate exceptional bursts of audience attention, making breakout content an important KPI for the channel.

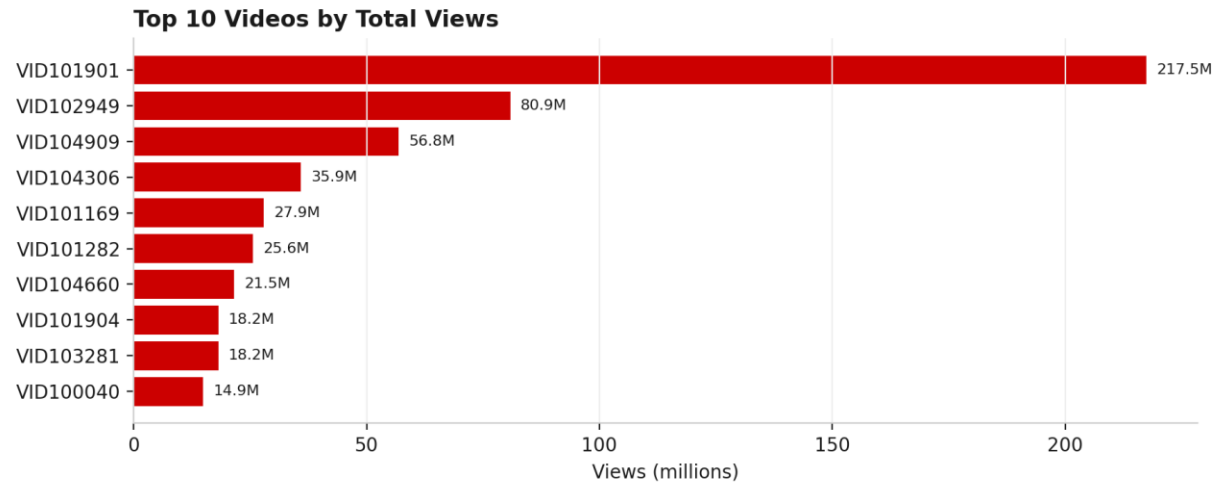
4. Top-Performing Videos (By Views)

The top performer drew **217.5M** views — more than 2.5x the second-place video — a common "viral hit" pattern on the platform (a small number of uploads drive a disproportionate share of total viewership, a Pareto effect worth flagging for content and promotion strategy).

The graph shows that Music has the highest views, with over **217** million views, while People & Blogs has the highest likes and a very large subscriber base. Overall, Music, Sports, and People & Blogs are among the most popular categories, showing strong audience engagement and reach

Video ID	Category	Views	Likes	Comments	Channel Subscribers
VID101901	Music	217,500,554	16,814,107	1,538,810	339
VID102949	People & Blogs	80,855,529	2,191,214	491,602	50,000,000
VID104909	Sports	56,845,857	1,858,455	232,751	6,908
VID104306	Education	35,922,334	1,257,077	197,220	50,000,000
VID101169	Entertainment	27,945,776	1,190,267	248,727	9,690,335
VID101282	Sports	25,623,164	899,482	126,420	23,171,818
VID104660	Sports	21,549,234	1,056,744	150,813	10,736,346

Video ID	Category	Views	Likes	Comments	Channel Subscribers
VID101904	Science & Technology	18,198,722	1,039,704	189,289	22,748,145
VID103281	Gaming	18,165,065	1,509,974	141,502	34,583,888
VID100040	Music	14,904,085	450,755	41,813	37,300,780



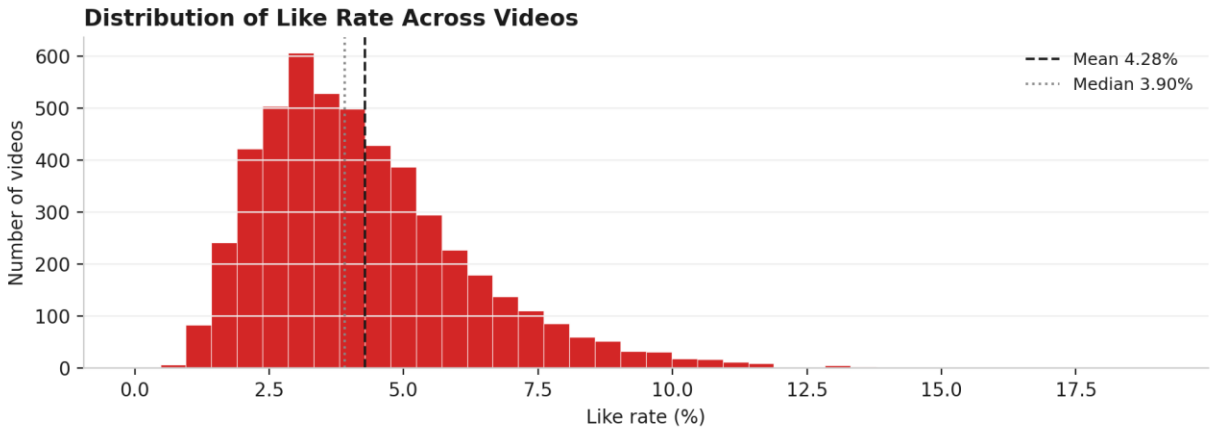
VID101901 is the clear standout performer, while **VID102949** and **VID104909** also have strong audience engagement. The sharp difference in views indicates that only a few videos achieve extremely high popularity.

5. Engagement Rate Distribution

Average like rate (likes ÷ views) across all 5,000 videos: 4.28%, with a median of 3.90% — moderately right-skewed, with the middle 50% of videos falling between 2.84% and 5.29%, and a small tail of videos reaching up to 19.0%. Average comment rate is a much thinner 0.63%, median 0.53%.

In a real system, this view is most useful segmented, not aggregated:

- **Videos with high views but bottom-quartile like rate —** 308 of the 5,000 tracked videos combine top-quartile viewership (108,000+ views) with a like rate below 2.84%, the highest-risk combination, since reaching amplifies the damage of weak audience reception.
- **Channel- or category-level rolling engagement trend —** the earliest signal of a content-fit or algorithm-distribution issue, best tracked as a moving average once a continuous feed replaces this snapshot.



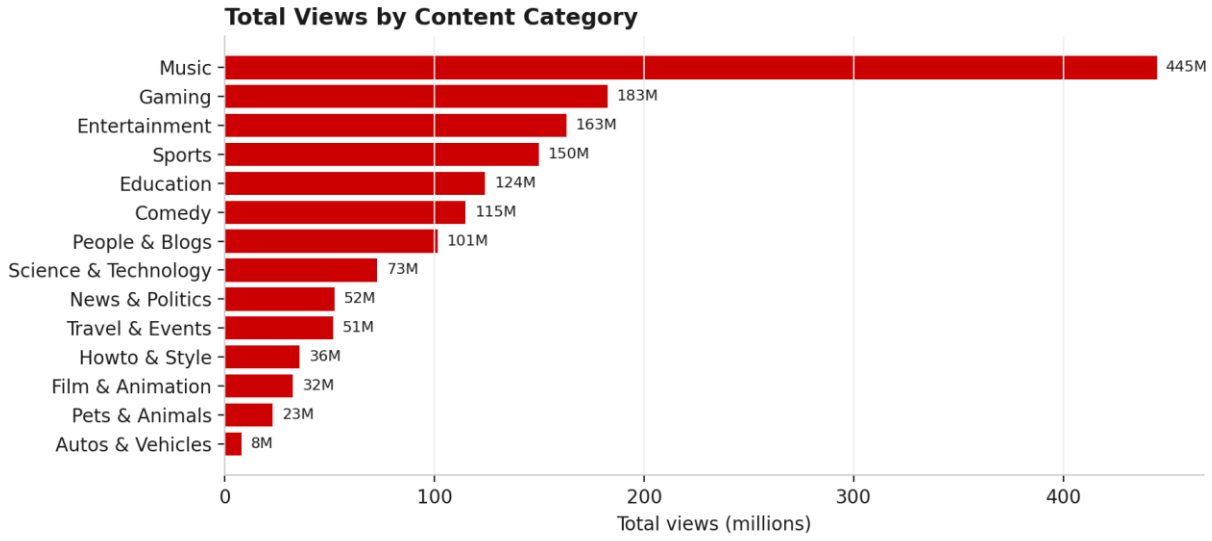
6. Category-Wise Demand

Viewer demand is heavily concentrated: Music leads with ~445M cumulative views from 678 videos (655,838 average views per video), followed by Gaming (~183M views, 750 videos) and Entertainment (~163M views, 605 videos). This reflects real platform patterns — a handful of core content categories typically account for the majority of total watch time, with the rest long-tailed. This view is what drives:

- Content investment and creator-partnership prioritization by category
- Category-specific recommendation and promotion tuning

- Advertiser-demand forecasting (high-view, high-engagement categories command premium ad inventory)

Category	Videos	Total Views	Avg. Views/Video	Avg. Like Rate
Music	678	444,658,270	655,838	4.90%
Gaming	750	182,525,681	243,368	5.34%
Entertainment	605	162,985,112	269,397	4.33%
Sports	337	149,735,786	444,320	3.71%
Education	425	124,160,013	292,141	3.21%
Comedy	421	114,826,184	272,746	5.43%
People & Blogs	152	101,406,155	667,146	3.26%
Science & Technology	480	72,657,929	151,371	3.64%
News & Politics	255	52,221,001	204,788	2.18%
Travel & Events	229	51,493,738	224,863	4.51%
Howto & Style	283	35,572,508	125,698	3.79%
Film & Animation	183	32,405,048	177,077	4.26%
Pets & Animals	97	22,840,191	235,466	5.88%
Autos & Vehicles	105	7,900,504	75,243	3.07%



KPI Dashboard Structure

KPI 1: Total Views

Formula: Total Views = SUM(Views)

Status	Target
Poor	**< 50M**
Average	**50M – 100M**
Good	**100M – 250M**
Excellent	**> 250M**

KPI 2: Like Rate

Formula**:** Like Rate =2(SUM(Likes) / SUM(Views)) × 100

Performance	Like Rate
Poor	**< 3%**
Good	**3% - 5%**
Excellent	**> 5%**

KPI 3: Engagement Rate

This is the most important KPI.

Formula: Engagement Rate =2((Likes + Comments) / Views) × 100

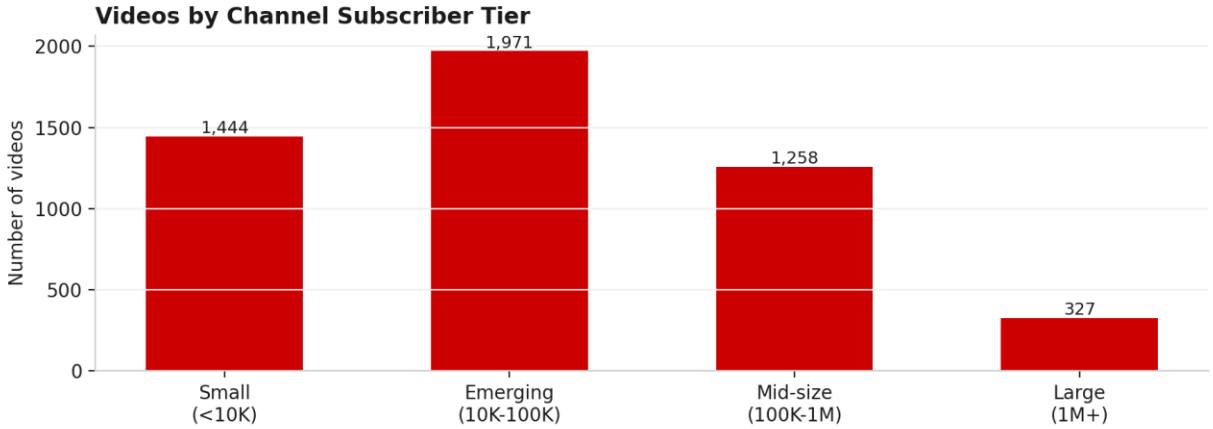
Performance	Value
Poor	< 4%
Good	4% - 6%
Excellent	> 6%

7. Channel Reach Status

Videos were bucketed by their channel's subscriber count into four tiers:

- **Large (1M+ subscribers) —** 327 videos, averaging 2.33M views each — established channel reach.
- **Mid-size (100K–1M) —** 1,258 videos, averaging 236K views each.
- **Emerging (10K–100K) —** 1,971 videos, averaging 75K views each.
- **Small (<10K subscribers) —** 1,444 videos (28.9% of the dataset) — the tier most exposed to algorithmic under-distribution, should be cross-referenced against the top-performers list above; a small channel already producing a breakout hit is a materially bigger growth opportunity than a large channel doing the same.

352 videos combine top-quartile views (108,000+) with a small-or-emerging channel (under 100K subscribers) — breakout content that is outperforming its channel's existing reach, the clearest signal in this dataset for creator-partnership or promotional-boost prioritization.



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8. Summary of Key Metrics

Metric	Value
Total videos tracked	5,000
Date range	Jan 2023 – Aug 2026 (43 months)
Total cumulative views	~1.56B
Total likes	~75.5M
Total comments	~10.3M
Average like rate	4.28% (median 3.90%)
Average comment rate	0.63% (median 0.53%)
Top single video (views)	217,500,554
Content categories tracked	14
Videos from small/emerging channels (<100K subs) that broke into the top view quartile	352 of 5,000

9. Next Steps to Move from Descriptive → Predictive

This report answers what happened. The natural next layer, if useful:

- **View and engagement forecasting per video/category to guide upload scheduling and ad-inventory planning**
- **Anomaly detection on sudden engagement-rate drops or unexpected view spikes**
- **Cohort/retention analysis on subscriber growth and repeat-viewership behavior, not just raw view counts**

Happy to build out any of these next, or connect a live feed (YouTube Analytics API, a database export, etc.) if available — the same pipeline and chart structure above would apply directly.